

Extreme Precipitation Risk in Beijing (1960–2024): An EVT-Based Assessment

Liu, Felix

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Extreme Rainfall Disasters in Beijing

2012 Beijing “7.21” Torrential Rainstorm



2023 Beijing-Tianjin-Hebei rainstorm disaster



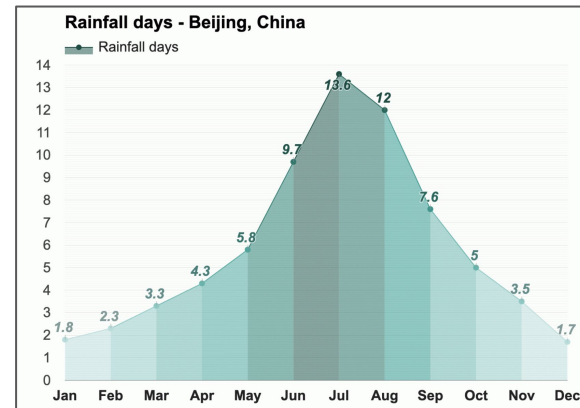
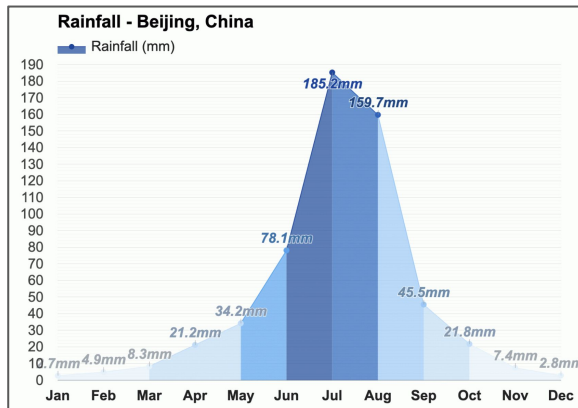
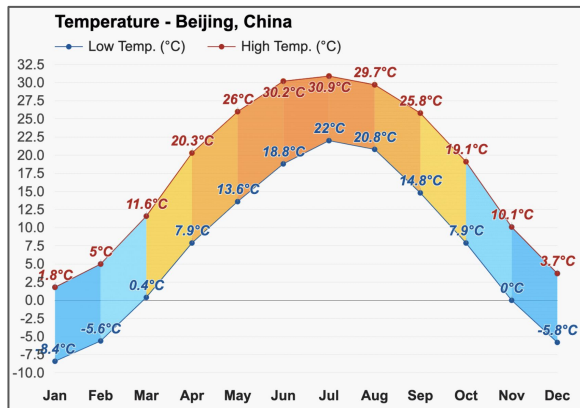
Image source: Caixin (2012); China News Service (2012, 2023); ABC News (2023)

Beijing's Seasonal Climate Pattern

 July: Hottest & Wettest Month

 185.2 mm average rainfall in July

 17 July 13.6 rainy days in July



 **High flood risk concentrated in midsummer (Jun–Aug)**

Recent Floods Highlight the Urgent Need for Risk Assessment



 **2012 “7-21” Rainstorm:** 79 deaths, urban traffic collapse

 **2023 Mentougou Flood:** >580 mm rainfall, 30,000+ evacuated

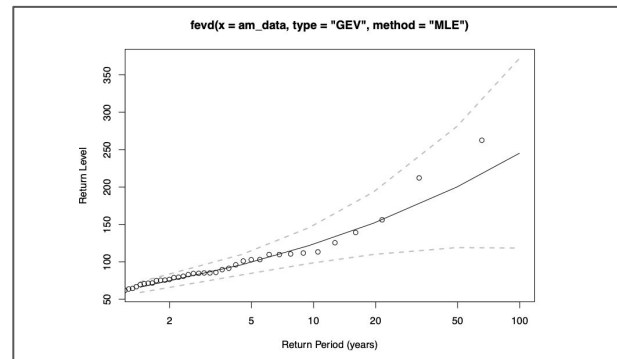
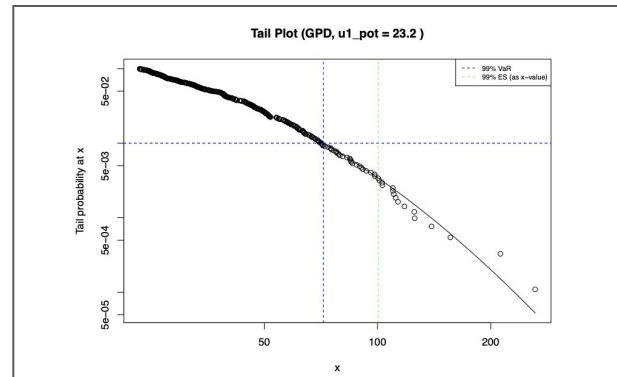
 **Flood severity not always linked to rainfall amount**

 **Traditional design standards may underestimate real risk**

 **Need for quantitative tools to assess rare but severe events**

Why Use Extreme Value Theory in Rainfall Analysis?

- EVT focuses on the tails of a distribution — the rare but impactful events
- Traditional models underestimate extreme precipitation probabilities
- Widely used in hydrology, insurance, and climate risk research
- Two main approaches:
 - **Annual Maxima (GEV)**
 - **Peaks Over Threshold (GPD)**



Previous Studies on Rainfall Extremes in Beijing

- **Ding et al. (2011)** Reviewed EVT advances in climate change research → emphasized heavy-tailed extremes
- **Song et al. (2018)** Compared AM-GEV vs. POT-GPD on 45 Beijing stations (1960–2012)
→ GEV yields more conservative return levels; GPD more sensitive to threshold choice
- **Song et al. (2019)** Linked urban flood risk to land-use changes, runoff dynamics & drainage infrastructure
- **He et al. (2020)** Used Bayesian MCMC with GEV/GPD to model extreme precipitation in Beijing, Shenzhen & Jinan → quantified parameter uncertainty
- **Miao & Chen (2008)** Demonstrated that Beijing's urban expansion intensifies summer short-duration heavy rainfall
- **Wu & Li (2017)** Developed non-stationary (time-varying) GEV models to capture trends in North China extremes

Approaches Used in the Literature

Study	Time Period	Method	Estimation	Notes
Song et al. (2018)	1960–2012	GEV & GPD	L-moments	45 stations in Beijing; return level maps
Song et al. (2019)	2000s onwards	Flood analysis	Observation	Linked rainfall risk to urbanization
Ding et al. (2011)	Various (review)	EVT overview	Various	Emphasis on climate relevance
He et al. (2020)	1961–2015	GEV/GPD + MCMC	Bayesian	Posterior inference via MCMC
Wu & Li (2017)	1960–2010	Non-stationary GEV	MLE	Shape/location modeled with time
This study	1960–2024	GEV & GPD	MLE (extRemes/qrmtools)	Updated to recent floods

Compared to earlier studies, this project extends the time range to include recent extreme events (e.g., 2023), and uses modern R packages to estimate risk metrics like return levels, VaR, and ES. While some studies used L-moments or Bayesian inference, this analysis applies Maximum Likelihood Estimation (MLE) for both GEV and GPD models.

Daily Rainfall in Beijing (1960–2024)

Data Source: NOAA GHCND station CHM00054511

Period: Jan 1, 1960 – Dec 31, 2024

Unit: Daily total precipitation (mm)

- Only **non-zero daily rainfall** was used for extreme value modeling
- The distribution is **right-skewed**, with a long tail of rare heavy rainfall events
- Highlights the need for **tail-focused statistical models** like EVT

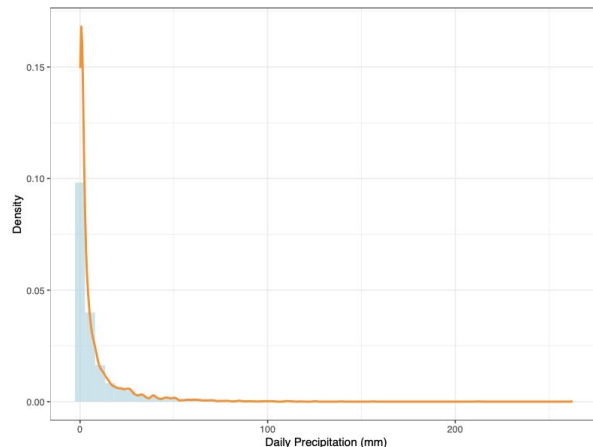


Figure 1: Daily Precipitation (mm) Distribution Plot

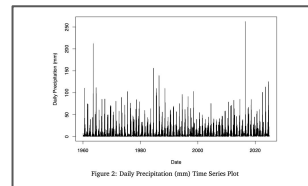
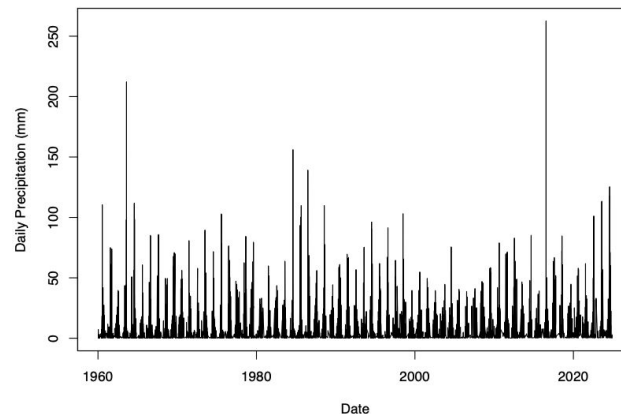
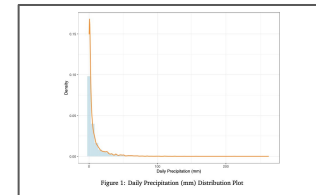


Figure 2: Daily Precipitation (mm) Time Series Plot

Daily Rainfall in Beijing (1960–2024)

- Daily rainfall shows **large interannual variability**
- Some years experience **much higher maximum rainfall** (e.g., 1963, 2012, 2016, 2023)
- **Extreme spikes** occur irregularly — not always in recent years
- These **rare high-impact events** are the main targets of risk modeling
- EVT is ideal for this type of **irregular and heavy-tailed behavior**



Extreme Value Theory Approaches: GEV vs. GPD

- **Block Maxima (GEV):**
Extracts the **maximum rainfall per year**
→ Fits a **Generalized Extreme Value (GEV)** distribution
→ Good for **return period estimation**
- **Peaks Over Threshold (GPD):**
Uses **all rainfall values above a chosen threshold**
→ Fits a **Generalized Pareto Distribution (GPD)**
→ More efficient use of tail data; sensitive to **threshold choice**
- Both models are applied to the **same dataset** for comparison
- Parameters estimated using **MLE**

Diagnostic Plots for GPD Threshold Selection

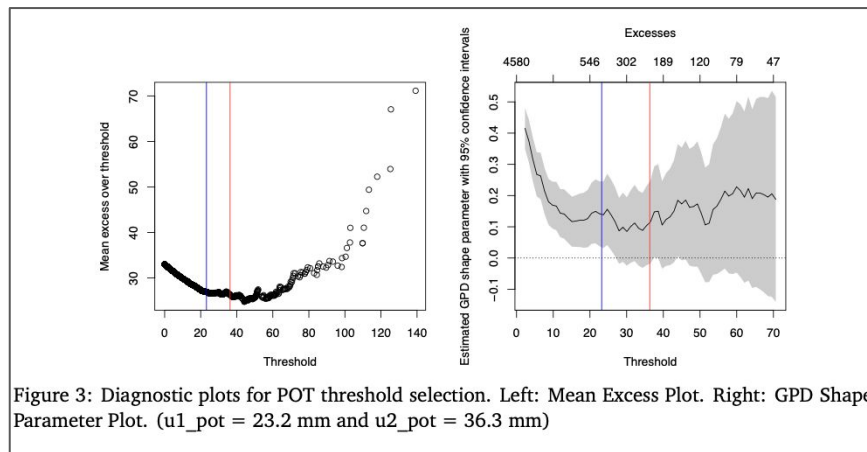
Mean Excess Plot (left):

- Look for **approximate linearity** above threshold
- stability starts around **23.2 mm**

Shape Parameter Plot (right):

- Tracks estimated **GPD shape ξ** as threshold varies
- Stability region between **23–36 mm**, with $\xi \approx 0.14$

- Threshold (u) is a critical choice in Peaks Over Threshold (POT) modeling
- Final threshold selected: **$u = 23.2$ mm**
 - Balances sample size and model stability



GPD Fit and Tail Risk Estimates (Threshold = 23.2 mm)

GPD model fitted to excesses above 23.2 mm using

Maximum Likelihood Estimation (MLE)

Estimated parameters:

- Shape (ξ) $\approx$ **0.14** $\rightarrow$ heavy-tailed
- Scale (σ) $\approx$ *17.9 mm*

Model convergence successful

Risk estimates (daily rainfall exceedance on wet days):

- **99% VaR** $\approx$ 71.7 mm
- **99% ES** $\approx$ 100.4 mm

These metrics indicate **high impact** even at rare exceedance levels

Table 2: GPD Model Fit Results for Threshold $u1_pot = 23.2$ mm (qrmtools MLE).

Parameter	Value
Shape (ξ)	0.1397
Scale (σ)	17.9036
Shape SE	0.0538
Scale SE	1.2713
Log-Likelihood	1835.2709
Convergence Status	Converged

Table 3: Estimated VaR and ES for GPD model with threshold $u1_pot = 23.2$ mm.

Level	VaR (mm)	ES (mm)
95%	36.14	59.05
99%	71.72	100.40

Return Levels Estimated from POT-GPD Model

Return level = rainfall amount **expected to be exceeded once every T years**

Based on POT-GPD fit with threshold = 23.2 mm

Return periods and levels:

- **10-year** → ~148 mm
- **50-year** → ~232 mm
- **100-year** → ~298 mm

These values are critical for **infrastructure design** and **urban flood protection**

Compared with real events: 2023 Mentougou flood

~580 mm ≈ above a 100-year event

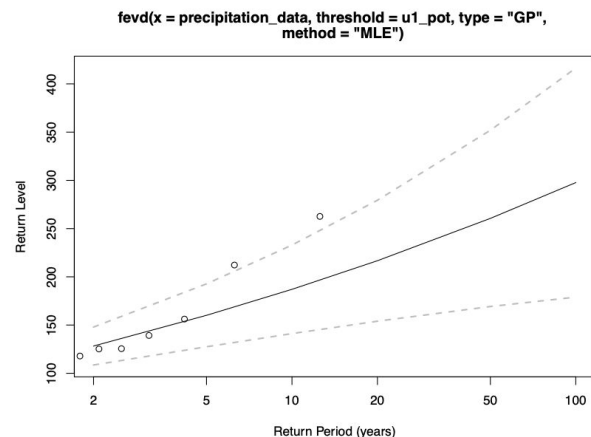


Figure 8: Return Level plot for the POT-GPD model ($u1_pot = 23.2$ mm).

Table 4: Estimated Return Levels (mm) for POT-GPD model ($u1_pot = 23.2$ mm).

Return Period (Years)	Return Level (mm)
10	187.02
50	260.56
100	297.69

GEV Return Level Analysis (Annual Maxima Method)

Based on the **Block Maxima (AM) method**

→ Uses **annual maximum daily rainfall** from 1960 to 2024

Model: **Generalized Extreme Value (GEV)**

→ Fitted using Maximum Likelihood Estimation (MLE)

Estimated parameters:

- Location (μ) ≈ 58.23 mm
- Scale (σ) ≈ 20.50 mm
- Shape (ξ) ≈ 0.27 → heavy tail

Return levels:

- **10-year** → ~121.7 mm
- **50-year** → ~199.9 mm
- **100-year** → ~245.1 mm

Table 5: GEV Model Fit Results for Annual Maxima data (extRemes MLE).

Parameter	Value
Location (μ)	58.2341
Scale (σ)	20.4993
Shape (ξ)	0.2697
Location SE	2.9896
Scale SE	2.4880
Shape SE	0.1249
Log-Likelihood	-308.8100
AIC	623.6201
BIC	630.1432
Convergence Status	Converged

Table 6: Estimated Return Levels (mm) for AM-GEV model.

Return Period (Years)	Return Level (mm)
10	121.68
50	199.94
100	245.05

Comparing Return Levels: GPD vs. GEV

Interpretation:

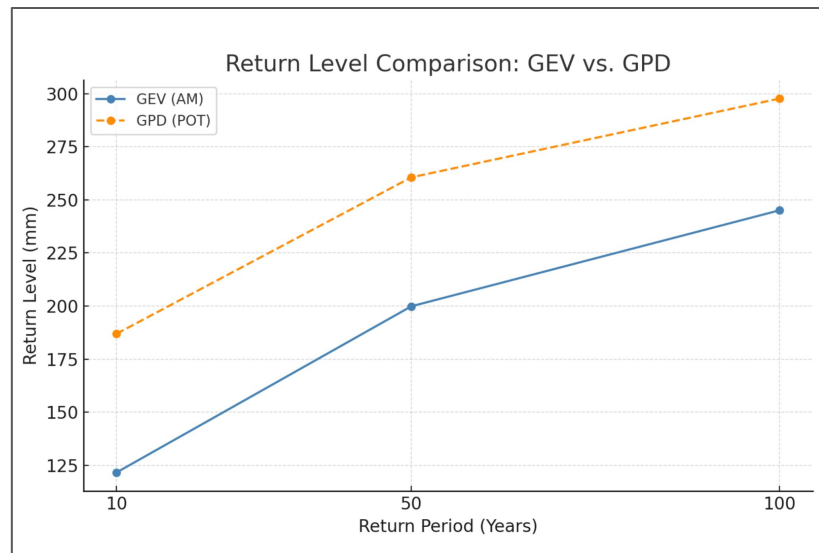
POT model yields **higher return levels** → more conservative for risk planning

GEV is more **stable** and widely used in engineering standards

Together, they provide a **range of estimates** that inform robust decision-making

Method Difference:

- POT-GPD: uses all daily rainfall **above threshold**
- AM-GEV: uses **one maximum** per year



Conclusion and Future Work

What We Learned

- Extreme rainfall events in Beijing are **irregular but increasing in severity**
- Both **GEV** and **GPD** models suggest **heavy-tailed risk**, confirming need for advanced modeling
- **POT-GPD estimates higher return levels**, offering a more conservative view for infrastructure design
- 100-year event magnitude (245–298 mm) is **within range of recent disasters (e.g., 2023)**

Future Work

- **Incorporate non-stationary EVT**
 - Capture potential trends due to **urbanization** or **climate change**
- Explore **Bayesian methods (e.g., MCMC)** to better assess parameter uncertainty
- Extend analysis to **multiple cities or stations** for regional comparison
- Integrate results into **flood planning models** and **urban drainage design**

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